



MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : MC 401/MC-401/MC401 Environmental Sciences

UPID : 004421

Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer any ten of the following :

[1 x 10 = 10]

- (i) What are the differences between Species, Population and Ecological Community ?
- (ii) What are the differences between Pollutants and Contaminants ?
- (iii) What are the main river water pollutants ? What is the range of Dissolved Oxygen concentration for healthy water ?
- (iv) What is Lithosphere and what are its major components ?
- (v) Which Smog is called "London Smog" and "Los Angeles Smog" ?
- (vi) How many main nutrients do we have in our environment ? Name them.
- (vii) What is a steady state system ? Indicate by means of mathematical term.
- (viii) What are the differences between Food Chain and Food Web ?
- (ix) What is Ozone layer depletion ? What is the full form of CFC and what are sources of CFCs ?
- (x) What are the full form of BOD and DO ? What are the most common heavy metals found in water ?
- (xi) What are the sources of Solid Waste ? What do you understand by 'Resource Recovery' from solid waste ?
- (xii) What is an aquifer ? How is water available in an aquifer ?

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 x 3 = 15]

2. (i) What is Environmental Laws in India ?
(ii) What are the differences between Law and Act ? [5]
3. (i) What is environment and what are its functions ? [5]
(ii) What is the relationship among Man, Society and Environment ? Show by flow diagram.
(iii) What do you understand by sustainable development ?
4. (i) What do you understand by 'Biogeochemical Cycle' ? How is it important to the environment ? [5]
(ii) How many main nutrient cycle are available for maintaining balance of environment ? Name them.
5. (i) What is Smog and what are two distinct types of Smog ? [5]
(ii) What are the differences between the two different types of Smogs ?
(iii) What are CFCs and what are their sources ?
6. (i) How resource is recovered from solid waste ? [5]
(ii) What is land filling of solid waste ?
(iii) What is composting ?

Group-C (Long Answer Type Question)

Answer any three of the following :

[15 x 3 = 45]

7. (i) What do you understand by Air Quality standard ? [15]
(ii) How many types of air quality standards do we have? Name them.
(iii) Why do we need air quality standards ?
(iv) What do you understand by ambient air quality standards ?
8. (i) What are water hardness, alkalinity and water softening ? [15]
(ii) What are toxic pollutants in water and what are their sources ?
(iii) Is Arsenic a toxic pollutant and how Arsenic is found in groundwater ?
9. (i) What are the effects of Suspended Particulate Matters ? [4 + 4 + 3 + 4]
(ii) What are effects of oxides of Nitrogen ?
(iii) Which oxide of Nitrogen is called laughing gas ?
(iv) What are the two important oxides of Sulphur ?

10. (i) What is Noise Pollution ? [3 + 3 + 4
(ii) What are the main sources of Noise Pollution ? +5]
(iii) What is the formula to calculate sound pressure ?
(iv) Determine the sound pressure level, if the rms pressure value is $30 \mu\text{Pa}$.
11. (i) What is Environmental Impact Assessment ? [3 + 4
(ii) What are various steps of EIA process ? +5+3]
(iii) What is Environmental Auditing and its purposes ?
(iv) Name five important Environmental Acts of India.

*** END OF PAPER ***